

Week 10: Magnetic Circuits and Materials

Magnetic field, flux, Faraday and Ampère, reluctance, air gaps, inductance, B–H curves, boundary conditions, magnetic energy

In-depth notes covering all core material of this chapter: theory, engineering shortcuts, exam traps, 8 worked examples, and 8 practice problems with full solutions.

Based on: Rizzoni & Kearns, *Principles and Applications of Electrical Engineering*, 7th ed., §14.1–14.3

Inspired by: ELEC 275, Principles of Electrical Engineering, Concordia University

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How to use this deck

- **“Try it” slides:** the problem alone, with empty space. Do it first.
- **Next slide:** the clean solution, step by step.
- **Purple boxes:** engineering shortcuts, the fast way to see or solve things.
- **Orange boxes:** traps. This is where marks get lost on exams.
- Formulas are tagged **[on formula sheet]** or **[NOT on formula sheet]**, based on the sheet given in ELEC 275 exams.
- **End of deck:** final-check problems, then a full answer sheet.

The right side of each slide is left empty for your notes.

Symbols and units

Symbol	Name	Unit	Plain meaning
H	field intensity	A/m	the “push” created by current
B	flux density	T = Wb/m ²	how concentrated the field is
$\mu = \mu_r \mu_0$	permeability	H/m	how easily a material carries flux: $B = \mu H$
μ_0	free space	$4\pi \times 10^{-7}$ H/m	air and vacuum
μ_r	relative permeability	–	iron: hundreds to thousands; air: 1
Φ	flux	Wb	total field through an area, $\Phi = BA$
λ	flux linkage	Wb-t	$\lambda = N\Phi$
$\mathcal{F}$	magnetomotive force	A-t	$\mathcal{F} = NI$, the magnetic “voltage”
$\mathcal{R}$	reluctance	A-t/Wb	the magnetic “resistance”, $\ell/(\mu A)$

Ampère's law and Faraday's law

[NOT on formula sheet: memorize]

Ampère: current makes field

$$\oint \mathbf{H} \cdot d\mathbf{l} = NI$$

Field $\times$ path length = current enclosed. Constant H along length ℓ :
 $H\ell = NI$.

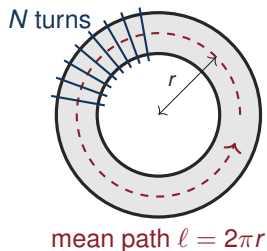
Faraday: changing flux makes voltage

$$e = \frac{d\lambda}{dt} = N \frac{d\Phi}{dt}$$

Only a **changing** flux induces a voltage. This is how transformers, generators and inductors work.

Gauss for magnetism: $\oint \mathbf{B} \cdot d\mathbf{S} = 0$. Flux lines always close (no monopoles): flux in = flux out of any region, like KCL.

The toroid: the standard example



Ampère $H = \frac{NI}{2\pi r}$

Flux density $B = \mu H$

Flux $\Phi = BA$ (A = cross-section)

Linkage $\lambda = N\Phi$

Inductance $L = \frac{\lambda}{I} = \frac{\mu N^2 A}{2\pi r}$

Engineering shortcut

$L \propto N^2$: doubling the turns **quadruples** the inductance. Doubling the area only doubles it. Adding turns is usually cheaper than adding iron.

Try it: Example 1, toroid

A toroid has $N = 500$ turns, mean radius $r = 10$ cm, cross-section $A = 4$ cm², $\mu_r = 500$, and carries $I = 2$ A.

Find H , B , Φ , λ and L .

your work here ↓

Example 1: solution

ℓ mean path of a toroid is its mean circumference: $2\pi(0.1) = 0.628$ m

H Ampère: $H = NI/\ell = \frac{500 \times 2}{0.628} = 1592 \text{ A/m}$

B $B = \mu_r \mu_0 H = 500(4\pi \times 10^{-7})(1592) = 1.0 \text{ T}$

Φ BA with $A = 4 \text{ cm}^2 = 4 \times 10^{-4} \text{ m}^2$: 0.4 mWb

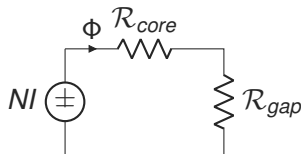
λ the flux links all 500 turns: $N\Phi = 500 \times 0.4 \times 10^{-3} = 0.2 \text{ Wb-t}$

L $\lambda/I = 0.2/2 = 0.1 \text{ H}$ (check: $\mu N^2 A/\ell = 0.1 \text{ H} \checkmark$)

Magnetic circuits: Ohm's law for flux

[NOT on formula sheet: memorize]

Electric circuit	Magnetic circuit	
voltage V	MMF $\mathcal{F} = NI$	(A-turns)
current I	flux Φ	(Wb)
resistance $R = \ell/(\sigma A)$	reluctance $\mathcal{R} = \ell/(\mu A)$	(A-t/Wb)
Ohm's law $V = IR$	$\mathcal{F} = \Phi \mathcal{R}$	
KVL, KCL	the same rules for MMF and flux	



A core with an air gap is a source (NI) driving the core and gap reluctances in series.

Trap: this assumes constant μ (no saturation) and uniform B . For a nonlinear core, use the B-H curve (later).

Why reluctance acts like resistance

Follow a core of length ℓ and cross-section A with N turns carrying I :

Ampère along the core $H\ell = NI \Rightarrow H = NI/\ell$

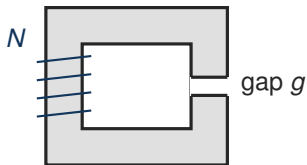
Material $B = \mu H = \mu NI/\ell$

Flux $\Phi = BA = \frac{\mu A}{\ell} NI$

Rearrange $NI = \Phi \frac{\ell}{\mu A} = \Phi \mathcal{R}$

That's exactly $V = IR$ with MMF as the “voltage”, flux as the “current” and $\mathcal{R} = \ell/(\mu A)$ as the “resistance”. Like a wire, a long thin path of poor material has high reluctance. **Why series reluctances add:** the same flux passes through every part of one loop. Each part uses $H_k \ell_k = \Phi \mathcal{R}_k$ of the MMF, and Ampère says these add to NI .

Air gaps dominate



Air has $\mu_r = 1$. Iron has μ_r in the thousands. So a 1 mm gap can have **more** reluctance than 50 cm of iron.

Reluctances in series add:

$$\mathcal{R} = \mathcal{R}_{core} + \mathcal{R}_{gap}.$$

Engineering shortcut: gap-only estimate

If $\mathcal{R}_{gap} \gg \mathcal{R}_{core}$: $NI \approx (B/\mu_0) g$. Usually within a few percent (P8).

Trap: the gap uses μ_0 (not $\mu_r\mu_0$), and the core length is the mean path **minus** the gap.

How much a gap dominates

Same area A for gap and core, so

$$\frac{\mathcal{R}_{gap}}{\mathcal{R}_{core}} = \frac{g/(\mu_0 A)}{\ell_{core}/(\mu_r \mu_0 A)} = \frac{\mu_r g}{\ell_{core}}$$

Example 2's core: $\frac{2000 \times 1 \text{ mm}}{500 \text{ mm}} = 4$: the 1 mm gap has **4 times** the reluctance of 50 cm of iron.

Physically: iron's domains line up and “help” the flux, so it needs very little H . Air has nothing to help, so almost all the MMF is used up pushing flux across the gap.

Rule of thumb: if $\mu_r g \gg \ell_{core}$, you can ignore the iron and use $NI \approx (B/\mu_0) g$.

Try it: Example 2, core with an air gap

A core has mean length 0.5 m (iron part), $\mu_r = 2000$, cross-section 10 cm^2 , and an air gap of 1 mm. The coil has $N = 200$ turns.

Find the current needed for $\Phi = 1 \text{ mWb}$, and what share of the MMF the gap takes. Ignore fringing.

your work here ↓

Example 2: solution

Plan series magnetic circuit: core and gap reluctances add,
then $NI = \Phi \mathcal{R}_{total}$

$$\mathcal{R}_{core} \quad \frac{\ell}{\mu_r \mu_0 A} = \frac{0.5}{2000(4\pi \times 10^{-7})(10^{-3})} = 1.99 \times 10^5 \text{ A-t/Wb}$$

$(A = 10 \text{ cm}^2 = 10^{-3} \text{ m}^2)$

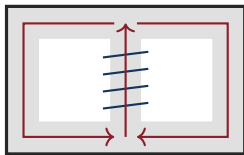
$$\mathcal{R}_{gap} \quad \text{air, } \mu_r = 1: \frac{10^{-3}}{(4\pi \times 10^{-7})(10^{-3})} = 7.96 \times 10^5 \text{ A-t/Wb}$$

$$\textbf{MMF} \quad \mathcal{F} = \Phi(\mathcal{R}_{core} + \mathcal{R}_{gap}) = 10^{-3} \times 9.95 \times 10^5 = 995 \text{ A-t}$$

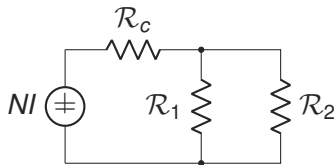
$$I \quad \mathcal{F} = NI: I = 995/200 = \boxed{4.97 \text{ A}}$$

Gap share $7.96/9.95 = \boxed{80\%}$ of the MMF goes to 1 mm of air

Parallel flux paths



shell core: coil on the centre leg



outer legs are in parallel

Parallel reluctances combine like parallel resistors, and flux splits like a current divider: the lower-reluctance leg takes more flux.

Try it: Example 3, shell core

Centre leg $\mathcal{R}_c = 1 \times 10^5$ A-t/Wb, each outer leg 2×10^5 A-t/Wb. The coil has $N = 100$ turns and $I = 2$ A.

Find the flux in the centre leg and in each outer leg.

your work here ↓

Example 3: solution

Draw MMF source (the coil on the centre leg) in series with $\mathcal{R}_c$, then the two outer legs in parallel

Parallel like resistors: $2 \times 10^5 \parallel 2 \times 10^5 = 1 \times 10^5 \text{ A-t/Wb}$

Total $\mathcal{R} = 1 \times 10^5 + 1 \times 10^5 = 2 \times 10^5 \text{ A-t/Wb}$

Centre $\Phi_c = \frac{NI}{\mathcal{R}} = \frac{100 \times 2}{2 \times 10^5} = 1 \text{ mWb}$

Outer flux divides like current; equal legs split it equally:
0.5 mWb each

Inductance from reluctance

[NOT on formula sheet: memorize]

$$L = \frac{N^2}{\mathcal{R}_{total}}$$

Why: $L = N\Phi/I = (N/I)(NI/\mathcal{R}) = N^2/\mathcal{R}$.

Example 2's core: $L = 200^2/(9.95 \times 10^5) = 40.2 \text{ mH}$

Engineering shortcut

A gap makes L smaller, but also makes it much more **stable**: the gap (air) is linear, so L barely changes as the iron nears saturation. That's why real inductors have gaps.

Magnetic materials

Type	Behaviour
Diamagnetic	no permanent dipoles; induced field opposes the applied field; μ_r slightly below 1 (copper, water)
Paramagnetic	weak permanent dipoles; induced field aligns with the applied field; μ_r slightly above 1 (aluminum)
Ferromagnetic	dipoles group into domains that align strongly; μ_r from hundreds to over 100 000 (iron, steel, nickel); can become permanent magnets

Only ferromagnetic materials matter for cores; the other two behave essentially like air ($\mu_r \approx 1$).

Where μ_r comes from: magnetization

[NOT on formula sheet: memorize]

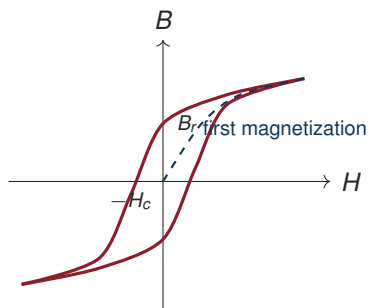
The atoms' aligned dipoles add their own field, the **magnetization** M , on top of H :

$$M = \chi_m H \quad B = \mu_0(H + M) = \mu_0(1 + \chi_m)H \quad \mu_r = 1 + \chi_m$$

χ_m is the **magnetic susceptibility**.

Material	χ_m	μ_r
Diamagnetic	small, negative	just below 1
Paramagnetic	small, positive	just above 1
Ferromagnetic	huge, not constant	hundreds to 10^5 (use the B–H curve)

B–H curves: saturation and hysteresis



Saturation: past a point, more H gives almost no more B . μ is **not** constant.

Hysteresis: B depends on the history. The loop's area is energy lost per cycle (heat).

Remanence B_r : flux left at $H = 0$ (permanent magnet).

Coercivity H_c : field needed to erase it.

Core losses

Hysteresis loss (loop area $\times$ frequency) and **eddy-current loss** (currents induced in the iron itself). Laminated cores (thin insulated sheets) cut eddy currents.

Try it: Example 4, using a B–H table

H (A/m)	B (T)
500	1.0
1000	1.4
2000	1.5

A toroid has mean radius 10 cm, $N = 500$ turns and cross-section 1 cm^2 . Its material follows this table.

- (a) Current that gives $H = 1000 \text{ A/m}$, and the flux.
(b) Double that current. By how much does the flux rise?

your work here ↓

Example 4: solution

(a) H Ampère: $H = NI/\ell \Rightarrow I = H\ell/N = \frac{1000 \times 2\pi(0.1)}{500} =$

1.26 A

(a) B nonlinear: read B from the table at $H = 1000$: $B = 1.4$ T

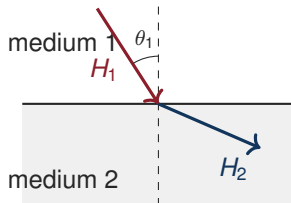
(a) Φ $BA = 1.4 \times 10^{-4} =$ 0.14 mWb ($A = 1 \text{ cm}^2 = 10^{-4} \text{ m}^2$)

(b) $2I$ gives $H = 2000$ A/m; the table says $B = 1.5$ T: flux 0.15 mWb, only +7%

Why the material is **saturating**: most domains are already aligned, so extra H barely adds B

Boundary conditions between two materials

[NOT on formula sheet: memorize]



At the surface between two materials:

Normal B is continuous: $B_{1n} = B_{2n}$
(from Gauss: flux can't disappear)

Tangential H is continuous: $H_{1t} = H_{2t}$,
if there is no surface current (from
Ampère)

So $\mu_1 H_{1n} = \mu_2 H_{2n}$.

Trap

Normal: match B , not H . Tangential: match H , not B . With a surface current K : $H_{1t} - H_{2t} = K$.

Why those two conditions

Normal B is continuous (from Gauss): imagine a flat pill box straddling the surface. Flux lines can't start or stop, so whatever flux enters one face leaves the other: $B_{1n}A = B_{2n}A$.

Tangential H is continuous (from Ampère): imagine a thin rectangle straddling the surface. Walk around it: $\oint \mathbf{H} \cdot d\mathbf{l} = H_{1t}\ell - H_{2t}\ell$. With no current enclosed that's zero, so $H_{1t} = H_{2t}$. A surface current K (A/m) makes the difference equal K .

How to use them: split the incoming field into normal and tangential parts, carry each across with its rule, and recombine.

Try it: Example 5, field entering iron

A field $H_1 = 10 \text{ A/m}$ in air hits a flat boundary at 30° from the normal, entering a material with $\mu_r = 4$. No surface current.

Find H_2 (magnitude and angle from the normal).

your work here ↓

Example 5: solution

Split normal part $H_{1n} = 10 \cos 30^\circ = 8.66$, tangential part $H_{1t} = 10 \sin 30^\circ = 5 \text{ A/m}$

Tangential carries straight across (no surface current): $H_{2t} = 5 \text{ A/m}$

Normal B_n carries across: $\mu_0 H_{1n} = 4\mu_0 H_{2n} \Rightarrow$
 $H_{2n} = 8.66/4 = 2.17 \text{ A/m}$

Recombine $|H_2| = \sqrt{5^2 + 2.17^2} = 5.45 \text{ A/m}$, angle from the
normal $\tan^{-1}(5/2.17) = 66.6^\circ$

Entering a higher- μ material, the normal part shrinks while the tangential part stays, so the field bends **away** from the normal.

Magnetic energy

[NOT on formula sheet: memorize]

$$W = \frac{1}{2}LI^2 \quad w = \frac{B^2}{2\mu} = \frac{1}{2}\mu H^2 \text{ (J/m}^3\text{)} \quad W = w \times \text{volume}$$

Engineering shortcut: most energy sits in the gap

For the same B , energy density is μ_r times bigger in air than in iron. In Example 2: gap volume 1 cm^3 stores 0.40 J , while 500 cm^3 of iron stores only 0.10 J . Total $0.50 \text{ J} = \frac{1}{2}LI^2 \checkmark$

Energy and power are how different engineering fields talk to each other. A magnet designer specifies the field; the power engineer sizes the supply from the energy needed.

Where $w = B^2/2\mu$ comes from

Take a toroid: $L = \mu N^2 A / \ell$ and $B = \mu NI / \ell$, so $I = B\ell / (\mu N)$. Its stored energy:

$$W = \frac{1}{2}LI^2 = \frac{1}{2} \cdot \frac{\mu N^2 A}{\ell} \cdot \frac{B^2 \ell^2}{\mu^2 N^2} = \frac{B^2}{2\mu} (A\ell)$$

$A\ell$ is the core's volume, so the energy **per cubic metre** is $w = \frac{B^2}{2\mu}$. Using

$B = \mu H$ it's also $\frac{1}{2}\mu H^2 = \frac{1}{2}BH$.

Why the gap stores more: at the same B , w is inversely proportional to μ . Air has $\mu_r = 1$ vs iron's thousands, so each cubic metre of gap holds thousands of times more energy.

Try it: Example 6, induced voltage

A 100-turn coil links a flux $\Phi(t) = 0.01 \sin(377t)$ Wb.

Find the induced voltage $e(t)$ and its peak and RMS values.

your work here ↓

Example 6: solution

Faraday $e = N d\Phi/dt$, and the derivative of $\sin(377t)$ is $377 \cos(377t)$

$$e(t) = 100 \times 0.01 \times 377 \cos(377t) = 377 \cos(377t) \text{ V}$$

Peak, RMS peak 377 V , $\text{RMS} = 377/\sqrt{2} = 267 \text{ V}$

Shortcut: for sinusoidal flux $E_{rms} = \frac{2\pi}{\sqrt{2}} f N \Phi_{max} = 4.44 f N \Phi_{max}$. Check:
 $4.44 \times 60 \times 100 \times 0.01 = 267 \text{ V} \checkmark$

Try it: Example 7, energy in a gapped core

Use Example 2 (gap 1 mm, cross-section 10 cm^2 , core volume 500 cm^3 , $\mu_r = 2000$, $B = 1 \text{ T}$).

Find the energy stored in the gap and in the iron, and check against $\frac{1}{2}LI^2$ ($L = 40.2 \text{ mH}$, $I = 4.97 \text{ A}$).

your work here ↓

Example 7: solution

Gap w $\frac{B^2}{2\mu_0} = \frac{1}{2(4\pi \times 10^{-7})} = 3.98 \times 10^5 \text{ J/m}^3$

Gap W volume $= A \times g = 10^{-3} \times 10^{-3} = 10^{-6} \text{ m}^3$: 0.398 J

Iron w $\frac{B^2}{2\mu_r\mu_0} = \frac{1}{2(2000)(4\pi \times 10^{-7})} = 199 \text{ J/m}^3$

Iron W volume $5 \times 10^{-4} \text{ m}^3$: 0.0995 J

Check $\frac{1}{2}LI^2 = \frac{1}{2}(0.0402)(4.97)^2 = 0.497 \text{ J} \approx 0.398 + 0.0995 \checkmark$

Meaning 80% of the energy sits in 0.2% of the volume: the gap

Self and mutual inductance

[NOT on formula sheet: memorize]

Two windings N_1 and N_2 on the same toroid (mean length ℓ , area S):

Self $L_1 = \mu N_1^2 S / \ell$, $L_2 = \mu N_2^2 S / \ell$ flux through **itself**

Mutual $M = \mu N_1 N_2 S / \ell$ flux through the **other** coil ($M_{12} = M_{21}$)

Coupling $M = k \sqrt{L_1 L_2}$ $k = 1$ when all flux links both coils

Voltages $v_1 = L_1 di_1 / dt$, $v_2 = M di_1 / dt$ (coil 2 open)

Link to week 8: with $k = 1$, $v_2 / v_1 = M / L_1 = N_2 / N_1$: exactly the ideal transformer ratio.

Try it: Example 8, mutual inductance

A toroid (mean radius 5 cm, cross-section 2 cm^2 , $\mu_r = 1000$) carries $N_1 = 200$ and $N_2 = 50$ turns.

(a) Find L_1 , L_2 and M . (b) i_1 rises at 100 A/s with coil 2 open. Find v_1 , v_2 and their ratio.

your work here ↓

Example 8: solution (1/2)

Common factor every inductance has $\mu S/\ell$: $\ell = 2\pi(0.05) = 0.314$
m, $\frac{1000(4\pi \times 10^{-7})(2 \times 10^{-4})}{0.314} = 8 \times 10^{-7}$

$$L_1 \times N_1^2: 8 \times 10^{-7} \times 200^2 = \boxed{32 \text{ mH}}$$

$$L_2 \times N_2^2: 8 \times 10^{-7} \times 50^2 = \boxed{2 \text{ mH}}$$

$$M \times N_1 N_2: 8 \times 10^{-7} \times 200 \times 50 = \boxed{8 \text{ mH}}$$

Example 8: solution (2/2)

Check $\sqrt{L_1 L_2} = \sqrt{0.032 \times 0.002} = 8 \text{ mH} = M$, so $k = 1$ ✓

(b) $v_1 = L_1 di_1/dt = 0.032 \times 100 = 3.2 \text{ V}$;

$$v_2 = M di_1/dt = 0.008 \times 100 = 0.8 \text{ V}$$

Ratio $0.8/3.2 = 0.25 = N_2/N_1$ ✓ (the transformer ratio)

Traps

Check these before you box an answer

- Convert cm and cm^2 to m and m^2 first.
- $\mu = \mu_r \mu_0$: don't forget $\mu_0 = 4\pi \times 10^{-7}$.
- Air gap: $\mu_r = 1$. Core length: subtract the gap.
- MMF is NI (ampere-turns), not I .
- Nonlinear core: use the B–H table, not a constant μ .
- $L = N^2/\mathcal{R}$: N squared.
- Fringing (flux spreading at the gap) is ignored unless stated.

Exam-format traps

Draw the magnetic circuit

Draw the equivalent: an MMF source (NI) with reluctances in series or parallel. It organizes the problem and earns method marks.

Justify approximations

If you use the gap-only shortcut, say so and quote the ratio $\mathcal{R}_{gap}/\mathcal{R}_{core}$.

Summary

Basics

$$B = \mu H, \quad \Phi = BA, \quad \lambda = N\Phi, \quad \mu = \mu_r \mu_0$$

Laws

Ampère $H\ell = NI$; Faraday $e = N d\Phi/dt$; flux lines close

Circuits

$\mathcal{F} = NI = \Phi\mathcal{R}$, $\mathcal{R} = \ell/(\mu A)$; series add, parallel like resistors

Gaps

dominate reluctance and store most of the energy

Inductance

$$L = N\Phi/I = N^2/\mathcal{R}$$

Materials

dia, para ($\mu_r \approx 1$), ferro (domains, saturation, hysteresis);
use B–H data when nonlinear

Boundaries

B_n continuous, H_t continuous (no surface current)

Energy

$$\frac{1}{2}LI^2 = \int \frac{1}{2}B^2/\mu \, dV; \quad E_{rms} = 4.44fN\Phi_{max}$$

Final check

8 problems. One per slide, with space. No looking back.

If one takes more than 10 min, go back to that section. The answer sheet comes after the last problem.

P1: toroid

$N = 400$, $I = 1.5$ A, mean radius 8 cm, $\mu_r = 800$, cross-section 2 cm^2 .

Find H , B , Φ and L .

your work here ↓

P2: gapped core

Iron path 0.4 m, $\mu_r = 1500$, cross-section 5 cm², gap 0.5 mm, $N = 300$.

Find the current for $\Phi = 0.5$ mWb.

your work here ↓

P3: inductance

Find the inductance of the P2 core.

your work here ↓

P4: unequal legs

A shell core has centre leg $\mathcal{R}_c = 2 \times 10^5$, outer legs 3×10^5 and 6×10^5

A-t/Wb. $N = 200$, $I = 1$ A.

Find the flux in each leg.

your work here ↓

P5: B–H table

A toroid of mean radius 5 cm with $N = 200$ uses the same table as Example 4.

What current gives $H = 1000$ A/m, and what is B there?

your work here ↓

P6: boundary

$H_1 = 20 \text{ A/m}$ in air hits a boundary at 45° from the normal, entering $\mu_r = 10$.
No surface current.

Find $|H_2|$ and its angle from the normal.

your work here ↓

P7: induced voltage

A 50-turn coil links a sinusoidal flux of peak 2 mWb at 60 Hz. Peak and RMS induced voltage?

your work here ↓

P8: exam style

A core: iron path 0.3 m, $\mu_r = 3000$, cross-section 4 cm², air gap 2 mm, $N = 500$.

(a) Current for $B = 0.8$ T in the gap. (b) The gap's share of the MMF. (c) The current if the iron is ignored, and the error.

your work here ↓

Answer sheet

Full worked solutions to P1–P8.

P1: solution

$$H \quad \frac{NI}{2\pi r} = \frac{400 \times 1.5}{2\pi(0.08)} = \frac{600}{0.503} = 1194 \text{ A/m}$$

$$B \quad \mu_r \mu_0 H = 800(4\pi \times 10^{-7})(1194) = 1.2 \text{ T}$$

$$\Phi \quad BA = 1.2 \times 2 \times 10^{-4} = 0.24 \text{ mWb}$$

$$L \quad N\Phi/I = \frac{400 \times 0.24 \times 10^{-3}}{1.5} = 64 \text{ mH}$$

P2: solution

$$A \quad 5 \text{ cm}^2 = 5 \times 10^{-4} \text{ m}^2, \text{ gap } 0.5 \text{ mm} = 5 \times 10^{-4} \text{ m}$$

$$\mathcal{R}_{core} \quad \frac{0.4}{1500(4\pi \times 10^{-7})(5 \times 10^{-4})} = 4.24 \times 10^5 \text{ A-t/Wb}$$

$$\mathcal{R}_{gap} \quad \frac{5 \times 10^{-4}}{(4\pi \times 10^{-7})(5 \times 10^{-4})} = 7.96 \times 10^5 \text{ A-t/Wb}$$

$$\text{Total series: } 1.22 \times 10^6 \text{ A-t/Wb}$$

$$I \quad I = \frac{\Phi \mathcal{R}}{N} = \frac{0.5 \times 10^{-3} \times 1.22 \times 10^6}{300} = \boxed{2.03 \text{ A}}$$

P3: solution

Formula $L = N^2 / \mathcal{R}_{total}$ (from $L = N\Phi / I$ and $\Phi = NI / \mathcal{R}$)

Result
$$\frac{300^2}{1.22 \times 10^6} = \frac{90\,000}{1.22 \times 10^6} = 73.8 \text{ mH}$$

P4: solution

Parallel outer legs: $\frac{(3 \times 10^5)(6 \times 10^5)}{3 \times 10^5 + 6 \times 10^5} = 2 \times 10^5 \text{ A-t/Wb}$

Total add the centre leg: $2 \times 10^5 + 2 \times 10^5 = 4 \times 10^5$

Centre $\Phi_c = \frac{200 \times 1}{4 \times 10^5} = 0.5 \text{ mWb}$

Split flux divider (other reluctance on top): the 3×10^5 leg gets

$$0.5 \cdot \frac{6}{3 + 6} = 0.333 \text{ mWb}, \text{ the } 6 \times 10^5 \text{ leg gets}$$

$$0.167 \text{ mWb}$$

P5: solution

I Ampère: $I = \frac{H \cdot 2\pi r}{N} = \frac{1000 \times 2\pi(0.05)}{200} = 1.57 \text{ A}$

B nonlinear core: read the table at $H = 1000$: 1.4 T

P6: solution

Split at 45° : $H_{1n} = H_{1t} = 20 \cos 45^\circ = 14.14 \text{ A/m}$

Tangential carries across: $H_{2t} = 14.14 \text{ A/m}$

Normal B_n carries across: $H_{2n} = H_{1n}/\mu_r = 14.14/10 = 1.41 \text{ A/m}$

Result $\sqrt{14.14^2 + 1.41^2} = 14.2 \text{ A/m}$, $\tan^{-1}(14.14/1.41) = 84.3^\circ$ from the normal

P7: solution

Peak $e = N d\Phi/dt$ with $\Phi = \Phi_m \sin \omega t$: peak

$$= N\omega\Phi_m = 50 \times 377 \times 0.002 = 37.7 \text{ V}$$

RMS $37.7/\sqrt{2} = 26.7 \text{ V}$ ($= 4.44 \times 60 \times 50 \times 0.002 \checkmark$)

P8: solution

$$\Phi BA = 0.8 \times 4 \times 10^{-4} = 0.32 \text{ mWb}$$

$$\mathcal{R}_{core} = \frac{0.3}{3000(4\pi \times 10^{-7})(4 \times 10^{-4})} = 1.99 \times 10^5$$

$$\mathcal{R}_{gap} = \frac{0.002}{(4\pi \times 10^{-7})(4 \times 10^{-4})} = 3.98 \times 10^6$$

$$(a) \quad I = \frac{\Phi(\mathcal{R}_c + \mathcal{R}_g)}{N} = \frac{0.32 \times 10^{-3} \times 4.18 \times 10^6}{500} = \boxed{2.67 \text{ A}}$$

$$(b) \quad \text{gap share: } 3.98/4.18 = \boxed{95\%}$$

$$(c) \quad \text{gap only: } \frac{0.32 \times 10^{-3} \times 3.98 \times 10^6}{500} = \boxed{2.55 \text{ A}}, \text{ about}$$

5% low